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(57) Abstract:

The invention provides a system (100) for lip-sync authenticity detection using spatial, spectral, and deep-learning based feature fusion. The system acquires an input video, extracts frames at a fixed temporal rate, and constructs paired lip-region frames through face and lip detection, followed by standardized cropping. Spatial forensic features including correlation, pixel-difference, and edge-density values, together with spectral features derived from a two-dimensional Fast Fourier Transform, are computed from the paired frames. A convolutional neural network with five convolution blocks and an attention mechanism processes the same paired frame to generate deep feature scores and classification probabilities. A feature-fusion engine combines handcrafted and deep-learning features to compute a Lip-Audio Synchronization Consistency Index (LASCI) for determining REAL_SYNC or FAKE_DESYNC classifications. An explainability module generates heatmaps, FFT maps, feature plots, and reasoning outputs to support transparent forensic assessment of lip-sync integrity.

FORM 2

THE PATENTS ACT, 1970

(39 of 1970)

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The Patents Rules, 2003

COMPLETE SPECIFICATION

(see section 10)

SYSTEM FOR LIP-SYNC AUTHENTICITY DETECTION USING SPATIAL, SPECTRAL, AND DEEP-LEARNING BASED FEATURE FUSION

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The following specification particularly describes the invention and the manner in which it is to be performed.

FIELD OF THE INVENTION:

The present invention relates to the field of artificial intelligence, machine learning, and deep learning, specifically to audio-visual forensics and multimedia authentication. The invention further pertains to computer vision–based lip-sync authenticity detection, employing spatial analysis, spectral analysis using Fast Fourier Transform (FFT), and deep-learning models with attention mechanisms for evaluating synchronization integrity in video frames.

BACKGROUND:

Conventional technologies in audio-visual processing mainly address audio–video delay correction, lip-sync generation, or basic synchronization verification. Prior patents and research works focus on timestamp-based delay correction, hardware latency measurement, or the creation of synchronized lip movements for speech generation. Existing machine learning approaches primarily concentrate on lip-reading, audio reconstruction, or realistic lip animation, without providing a forensic mechanism for determining whether lip motions in video frames are authentic or manipulated.

However, these known systems do not provide forensic-grade lip-sync authenticity detection, nor do they assess manipulated, desynchronized, or deepfake-generated lip movements. Prior art lacks the ability to detect spatial inconsistencies, spectral abnormalities from generative model artifacts, or subtle micro-expression mismatches in paired video frames. Furthermore, existing methods do not integrate multi-feature spatial analysis, 2-dimensional FFT-based spectral forensics, and attention-driven deep-learning models into a unified detection framework.

A significant gap also exists in systems capable of evaluating lip-sync integrity without audio, particularly for muted videos, CCTV footage, or videos where audio has been removed or altered. Current solutions are not designed to compute a unified authenticity score that combines handcrafted features and deep-learning outputs for reliable classification.

Accordingly, there is a need for a comprehensive lip-sync authenticity detection system capable of analyzing spatial, spectral, and deep-learning features together, providing explainable forensic outputs and detecting frame-level manipulation with improved reliability.

OBJECTIVES OF THE INVENTION:

The primary objective of the present invention is to provide a system for reliable lip-sync authenticity detection using a unified framework that integrates spatial, spectral, and deep-learning–based feature fusion to classify paired video frames as REAL_SYNC or FAKE_DESYNC. The invention aims to overcome limitations of existing technologies that focus only on delay correction, lip animation generation, or basic alignment verification without assessing manipulation or frame-level inconsistencies.

Another objective of the invention is to implement a multi-feature spatial analysis mechanism, including correlation scoring, pixel-intensity difference evaluation, and edge-density assessment, enabling the detection of subtle visual mismatches that may not be identifiable through traditional inspection methods.

A further objective of the invention is to incorporate a frequency-domain analysis module utilizing 2D Fast Fourier Transform (FFT) to identify spectral irregularities caused by synthetic blending, generative artifacts, or cross-frame inconsistencies, thereby strengthening the forensic reliability of the system.

The invention also has the objective of providing a deep-learning–based paired-frame classification model employing convolutional blocks and an attention mechanism focused on lip-region micro-movements. This enables the system to learn discriminative temporal–visual patterns essential for authentic lip-sync detection.

Another objective is to compute a Lip-Audio Synchronization Consistency Index (LASCI) using a weighted fusion of spatial, spectral, and CNN-derived features, allowing the system to generate interpretable, graded mismatch severity levels for forensic applications.

A further objective of the invention is to enable authenticity detection even in silent-video scenarios, where audio is absent or removed, by relying solely on visual indicators extracted from paired frames. Additionally, the invention aims to provide an explainability and forensic reporting module that generates heatmaps, attention visualizations, FFT spectrum plots, feature comparisons, and reasoning statements to support transparent decision-making and legal admissibility

SUMMARY:

The present invention provides a lip-sync authenticity detection system that determines whether paired video frames represent synchronized or manipulated lip movements. The system implements a multi-stage analytical framework combining spatial features, spectral forensics, and deep-learning–based attention mechanisms to identify inconsistencies arising from desynchronization or artificial frame generation.

The invention extracts spatial indicators such as correlation, pixel-intensity differences, and edge-density variations from lip-region frame pairs to capture micro-expression discrepancies. In parallel, a frequency-domain module applies a 2D Fast Fourier Transform (FFT) to detect spectral abnormalities caused by synthetic blending or generative artifacts. A custom paired-frame convolutional neural network with an attention mechanism further analyzes the combined frame input to determine real or fake lip-sync patterns.

Outputs from these modules are integrated into a unified Lip-Audio Synchronization Consistency Index (LASCI), which produces a weighted authenticity score and severity level. The system additionally includes an explainability and forensic reporting module that generates heatmaps, spectral maps, feature comparisons, and reasoning summaries to support transparent evaluation. The invention is capable of detecting lip-sync mismatch even in silent video conditions, enabling application in scenarios where audio is unavailable or compromised

BRIEF DESCRIPTION OF THE DRAWINGS:

The foregoing summary, as well as the following detailed description of the invention, will be better understood when read in conjunction with the appended drawings. For the purpose of assisting in the explanation of the invention, there are shown in the drawings, embodiments that are presently preferred and considered illustrative. It should be understood, however, that the invention is not limited to the precise arrangements and instrumentalities shown therein the drawings:

FIG. 1 illustrates the overall system architecture for lip-sync authenticity detection.

FIG. 2 depicts the training and validation accuracy and loss curves,

FIG. 3 shows the confusion matrix representing classification performance across REAL_SYNC and FAKE_DESYNC categories,

FIG. 4 presents the feature distribution plots for correlation, pixel difference, and edge-density metrics

FIG. 5 illustrates the FFT spectrum comparison between REAL and FAKE samples

FIG. 6 provides example outputs of correct REAL and FAKE classifications,

REFERENCE NUMERALS:

100 Overall System for lip-sync authenticity detection

DETAILED DESCRIPTION OF THE INVENTION:

The present invention will now be described more fully hereinafter. For the purposes of the following detailed description, it is to be understood that the invention may assume various alternative variations and step sequences, except where expressly specified to the contrary. Thus, before describing the present invention in detail, it is to be understood that this invention is not limited to particularly exemplified systems or embodiments that may, of course, vary. The use of examples anywhere in this specification including examples of any terms discussed herein is illustrative only, and in no way limits the scope and meaning of the invention or of any exemplified term. Likewise, the invention is not limited to various embodiments given in this specification.

As used herein, the singular forms "a," "an," and "the" include plural reference unless the context clearly dictates otherwise. The term "and/or" means one or all of the listed elements or a combination of any two or more of the listed elements.

The terms “preferred” and “preferably” refer to embodiments of the invention that may afford certain benefits, under certain circumstances. However, other embodiments may also be preferred, under the same or other circumstances. Furthermore, the recitation of one or more preferred embodiments does not imply that other embodiments are not useful, and is not intended to exclude other embodiments from the scope of the invention.

When the term “about” is used in describing a value or an endpoint of a range, the disclosure should be understood to include both the specific value and endpoint referred to.

As used herein the terms “comprises”, “comprising”, “includes”, “including”, “containing”, “characterized by”, “having” or any other variation thereof, are intended to cover a non-exclusive inclusion

The invention operates by analysing paired video frames through a multi-stage pipeline that combines spatial forensics, spectral analysis, and deep-learning–based classification to determine lip-sync authenticity. The system first acquires an input video, extracts frames at a fixed rate, and generates paired samples consisting of either consecutive frames (REAL) or mismatched frames (FAKE). A face detector and lip-region extractor crop the relevant region of interest, and a paired 128×256 lip-frame image is constructed for analysis. The spatial feature module computes correlation, pixel-difference, and edge-density scores to capture micro-movement irregularities, while a 2D Fast Fourier Transform (FFT) evaluates spectral inconsistencies indicative of manipulation. In parallel, a custom CNN with five convolution blocks and an attention mechanism processes the paired frame to generate deep features and class probabilities. These spatial, spectral, and deep-learning outputs are fused to compute a unified Lip-Audio Synchronization Consistency Index (LASCI), which produces an authenticity score and mismatch severity level. A decision module then classifies the input as REAL_SYNC or FAKE_DESYNC, and an explainability module generates heatmaps, spectral plots, feature comparisons, and reasoning outputs to support transparent forensic interpretation.

Referring to figure. 1, The invention begins with an input acquisition and pre-processing pipeline in which the system receives an RGB video obtained from structured datasets such as the GRID dataset or any real-world source. Frames are extracted from the video at a fixed temporal rate, for example 10 frames per second, to maintain consistency in lip-motion progression. From these frames, the system constructs paired samples for subsequent analysis. When generating REAL samples, the system selects consecutive frames from the same video, while FAKE samples are produced by pairing non-consecutive frames or frames taken from different videos or speakers. A face detector, such as a Haar Cascade or YOLO-Face detector, identifies the facial region from each frame, and a secondary detector isolates the lower-face region of interest that contains the lips. This lip region is cropped into a standardized 128×128 pixel image to ensure uniformity of downstream features. Two such lip crops are placed side-by-side to construct a unified 128×256 paired frame that serves as the single input to both the convolutional neural network pathway and the handcrafted feature-extraction pathway.

The system processes this paired frame using a dual-path feature extraction framework. In the deep-learning path, a custom-designed convolutional neural network processes the input to extract spatial information relevant to micro-expressions, motion patterns, and lip deformation between the paired frames. The network contains five convolution blocks that progressively encode higher-order spatial features. An attention mechanism is integrated into the architecture to highlight discriminative regions, such as lip boundaries and motion gaps, thereby improving the detection of subtle visual discrepancies. The network produces a softmax probability output representing the likelihood of the paired frame belonging to the REAL_SYNC or FAKE_DESYNC category. In addition to these class probabilities, the final embedding or activation vector generated by the network is forwarded to the fusion module as a deep feature score.

Simultaneously, the handcrafted analytical pipeline extracts four interpretable features from the paired frame. A two-dimensional Fast Fourier Transform is applied to detect frequency-domain inconsistencies that may arise due to manipulation or frame mismatch. A correlation score is computed to assess the similarity between the two frames, where higher values indicate synchronized natural lip motion. A pixel-difference score is calculated by measuring absolute pixel-wise deviation, with high values suggesting abrupt transitions indicative of mismatched or manipulated content. An edge-density score is derived using Sobel filters to identify anomalies in edge distribution that may result from synthetic blending or generative artifacts. These handcrafted features are merged into a structured feature vector that encapsulates spatial and spectral deviations between the frame pair.

The invention uses a feature fusion and scoring layer in which the handcrafted feature vector and the CNN deep feature vector are combined into a single hybrid representation. This fusion module computes the Lip-Audio Synchronization Consistency Index (LASCI), a weighted metric introduced as one of the key novelties of the system.

$$\text{LASCI} = w_1 C + w_2 P + w_3 F - w_4 D - w_5 E$$

The LASCI score incorporates the correlation score C, pixel-difference score P, FFT consistency score F, edge-density deviation D, and CNN-based inconsistency score E, using learned or manually optimized weights $w_1 \ldots w_5$, expressed conceptually as a weighted combination of these parameters. The LASCI score quantifies the overall integrity and authenticity of the lip-synchronization exhibited in the paired frames.

The decision logic module receives both the LASCI score and the CNN prediction outputs and classifies the paired input into REAL_SYNC, representing authentic lip-synchronized content, or FAKE_DESYNC, representing manipulated or mismatched content. The module then directs REAL classifications to an authenticity output branch and FAKE classifications to a manipulated output branch.

The system further incorporates an explainability and forensic reporting module that generates interpretable visual outputs to support transparent decision-making. These outputs include activation heatmaps, attention visualizations, frame-difference heatmaps, FFT spectral plots, and bar-charts showing the computed features. A structured forensic report is produced, providing a human-readable explanation of why a given sample was classified as REAL or FAKE. This explainability component greatly enhances practical usability by allowing operators, auditors, or forensic analysts to understand the basis of each classification decision.

In an alternative embodiment within the scope of the invention, the paired-frame generator may utilize different temporal distances between frames to create REAL_SYNC and FAKE_DESYNC samples, while still operating on the same standardized 128×128 lip-region crops and maintaining the combined 128×256 paired-frame format. The invention may also be implemented using any equivalent face or lip-detection model, provided that it performs the disclosed role of isolating the lower-face region of interest for feature extraction. Similarly, the convolutional neural network architecture may be varied in depth or filter configuration, so long as it preserves the essential paired-frame processing, the five-block feature-extraction concept, and the attention mechanism that highlights lip-region micro-movements. The system may also employ learned or manually adjusted LASCI weighting values tailored for specific datasets or deployment environments, while still computing the LASCI metric from the same constituent features correlation, pixel difference, FFT consistency, edge-density deviation, and CNN inconsistency score. The explainability module may render the disclosed interpretability outputs in different visual formats or scaling resolutions, provided it continues to generate heatmaps, FFT spectral maps, frame-difference visualizations, feature comparisons, and classification reasoning as described in the invention

The invention offers significant advantages by integrating spatial, spectral, and deep-learning–based features into a unified lip-sync authenticity detection framework. Unlike conventional systems that focus primarily on delay correction, lip-reading, or lip-generation tasks, this invention introduces a comprehensive forensic approach that evaluates subtle temporal inconsistencies in lip motion. By using paired-frame analysis and handcrafted metrics such as correlation, pixel-intensity difference, and edge-density deviation, the system captures micro-movement disparities that would otherwise be undetectable through direct visual inspection or simple alignment checks. This strengthens the reliability of the system in identifying manipulated or desynchronized frames.

Another advantage of the invention lies in its use of a frequency-domain analysis module based on a two-dimensional Fast Fourier Transform. This enables the detection of high-frequency spectral abnormalities characteristic of synthetic blending, generative model artifacts, or frame stitching, thereby extending the forensic capability beyond what spatial features alone can reveal. The combination of spatial and spectral signals with attention-driven deep-learning features significantly enhances classification performance compared to methods relying on a single modality of analysis.

The introduction of the Lip-Audio Synchronization Consistency Index (LASCI) provides an additional advantage by generating a single, interpretable metric that quantifies authenticity using weighted contributions from all major feature sources. This allows the system to grade mismatch severity, making it suitable for forensic auditing, streaming-content verification, and authenticity scoring applications. The LASCI formulation ensures that the system remains both explainable and tunable for different deployment conditions, thereby improving adaptability and operational robustness.

A further advantage of the invention is the embedded explainability and forensic reporting module, which produces heatmaps, FFT spectral plots, feature bar charts, frame-difference maps, probability indicators, and reasoning summaries. These outputs provide transparent justification for each classification result, making the system suitable for legal, investigatory, and compliance-based use cases where interpretability is essential. The capacity to detect lip-sync inconsistencies even in silent videos offers another key benefit, enabling deployment in muted surveillance recordings, dubbing verification, and content-integrity monitoring scenarios where audio data may not be available or trustworthy.

Collectively, these advantages demonstrate that the invention not only improves detection accuracy relative to existing approaches but also enhances interpretability, robustness, and forensic value, making it highly suitable for real-world and investigative environments

Experimental Data:

The invention was experimentally validated using the GRID Lip-Reading Dataset, which contains high-quality audio-visual recordings of thirty-four speakers. A subset of videos from the dataset was processed to generate REAL_SYNC samples consisting of consecutive frames of the same video, and FAKE_DESYNC samples consisting of mismatched or non-consecutive frames. A total of sixty unseen paired samples were used for testing the system's performance.

The model was trained for five epochs using the constructed dataset of paired lip-region frames. Convergence was observed in both accuracy and loss values. The final training accuracy achieved by the model was approximately 63.57%, while the best validation accuracy recorded during the experiments was 70.00%. Validation loss consistently decreased over successive epochs, indicating stable learning behavior. The corresponding accuracy and loss curves (referred to as Fig. 2) showed progressive improvement in validation accuracy with a downward trend in loss.

A confusion matrix evaluation was performed on sixty test samples, consisting of thirty REAL and thirty FAKE samples. The experimental results demonstrated that twenty-seven out of thirty REAL samples were correctly classified, giving a REAL detection rate of 90%. Seventeen out of thirty FAKE samples were correctly classified. The overall test accuracy obtained was 73.3%. The precision, recall, and F1-score values for each class are reproduced below in their original form:

Class	Precision	Recall	F1-Score
REAL_SYNC	0.68	0.90	0.77
FAKE_DESYNC	0.85	0.57	0.68

The confusion matrix figure (Fig. 3) illustrated the count and percentage distribution of correct and incorrect classifications across both classes.

Feature-based validation was performed to assess the discriminatory power of handcrafted features such as correlation, pixel difference, and edge density. The correlation score for REAL samples ranged from approximately 0.95 to 1.00, whereas FAKE samples exhibited lower correlation values between approximately 0.82 and 0.90. Pixel-difference analysis showed that REAL samples typically produced values between 3 and 10, while FAKE samples showed substantially larger pixel differences ranging from 15 to 40. Edge-density evaluation further revealed that REAL samples generally produced values between 0 and 3, whereas FAKE samples exhibited higher edge-density values from 2 to 5. These distributions were illustrated in Fig. 4 and demonstrated clear separation between authentic and manipulated frames.

Frequency-domain validation was also performed using two-dimensional Fast Fourier Transform (FFT) magnitude spectra. Real samples showed consistent central-frequency concentration patterns, whereas fake samples exhibited irregular spectral patterns, cross-mixing distortions, and high-frequency noise components. The FFT spectrum comparisons referenced as Fig. 5 confirmed that spectral inconsistencies form a reliable indicator of content manipulation.

1. Sample-level case studies were conducted to analyze both correctly and incorrectly classified samples. Correct REAL classifications were associated with high lip-region correlation values (typically greater than 0.90), low pixel-intensity differences (less than 10), stable edge-density patterns, and uniform FFT spectral responses. These samples showed high LASCI scores exceeding the REAL threshold. Correct FAKE classifications were characterized by low correlation values (less than 0.85), high pixel-difference values (greater than 25), inconsistent lip structures or micro-expressions, higher edge-density variations, and spectral distortions in the FFT domain. The LASCI scores for these samples fell below the FAKE threshold. Representative examples of correct REAL and FAKE classifications, including their respective feature indicators, were presented in Fig. 6 .

The system also addressed cases where two frames visually appear to contain the same individual but are classified as DIFFERENT or FAKE_DESYNC due to temporal misalignment. Such occurrences arise from differences in timestamps, frame origin, or micro-expression continuity. The system correctly identifies mismatch by detecting reductions in correlation, increases in pixel intensity difference, and inconsistencies in lip-region micro-movements. This demonstrates the system's sensitivity to temporal alignment rather than mere facial identity.

A consolidated summary of system performance is reproduced exactly as disclosed:

Evaluation Parameter	**Result**
Test Accuracy	73.3%
Best Validation Accuracy	70.0%
Correct REAL Detection	90%
Correct FAKE Detection	56.7%
Interpretability Output	Provided
LASCI Score Correlation	Strong separation between REAL and FAKE

The experimental results collectively confirm that the hybrid architecture combining CNN attention features, handcrafted spatial features, and FFT-based spectral analysis delivers robust classification performance and strong interpretability. The LASCI metric showed clear separability between REAL and FAKE frame pairs, reinforcing the system's reliability for forensic lip-sync authentication.

WE CLAIM:

1. A system for lip-sync authenticity detection using spatial, spectral, and deep-learning based feature fusion (100), comprising:

   a paired-frame generator configured to construct a combined lip-region frame of size 128×256 pixels from two input frames obtained from an RGB video;

   a lip-region detection unit adapted to identify a lower-face region of interest and produce standardized 128×128 lip crops;

   a handcrafted feature extraction module configured to compute a correlation score, a pixel-difference score, an edge-density score, and a frequency-domain feature obtained through a two-dimensional Fast Fourier Transform applied to each lip crop;

   a convolutional neural network comprising five convolution blocks and an attention mechanism, the network being configured to extract deep spatial features and generate classification probabilities for REAL_SYNC and FAKE_DESYNC;

   a feature fusion engine adapted to combine the handcrafted features and the deep features;

   a Lip-Audio Synchronization Consistency Index (LASCI) computation unit configured to determine an authenticity score using weighted parameters C, P, F, D, and E, wherein C represents the correlation score, P represents the pixel-difference score, F represents the FFT consistency score, D represents the edge-density deviation, and E represents the CNN-based inconsistency score; and

   a decision logic module configured to classify the paired frame as REAL_SYNC or FAKE_DESYNC based on the LASCI score and the CNN output.

2. The system as claimed in claim 1, wherein the paired-frame generator is configured to generate REAL samples using consecutive frames from a same video and FAKE samples using non-consecutive or cross-video frames.
3. The system as claimed in claim 1, wherein the correlation score is computed as a quantitative measure of similarity between the paired lip crops, enabling detection of reduced temporal consistency in desynchronized content.
4. The system as claimed in claim 1, wherein the pixel-difference score is computed as an absolute pixel-wise deviation between the paired lip crops to detect abrupt visual transitions caused by frame swapping or manipulation.

5. The system as claimed in claim 1, wherein the edge-density score is derived using a Sobel-based edge estimator to identify irregular edge structures arising from synthetic blending or generative artifacts.
6. The system as claimed in claim 1, wherein the two-dimensional Fast Fourier Transform produces spectral magnitude representations that highlight high-frequency inconsistencies associated with manipulated frames.
7. The system as claimed in claim 1, wherein the attention mechanism of the convolutional neural network is configured to emphasize discriminative lip-region micro-movements by assigning higher weights to spatially relevant regions.
8. The system as claimed in claim 1, wherein the LASCI score is computed using learned or manually optimized weights w_1–w_5 to generate a fused authenticity measure derived from spatial, spectral, and deep-learning features.
9. The system as claimed in claim 1, wherein the decision logic module is further adapted to operate in silent-video conditions by utilizing only spatial, spectral, and deep-learning visual features without requiring audio input.
10. The system as claimed in claim 1, wherein an explainability and forensic reporting module is configured to generate activation heatmaps, attention visualizations, FFT spectral maps, frame-difference heatmaps, feature comparisons, and reasoning outputs that illustrate the basis for the classification decision..

Dated this 24th day of December 2025

Name and Signature of the Patent Agent
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For Daniel & Gladys.

ABSTRACT

SYSTEM FOR LIP-SYNC AUTHENTICITY DETECTION USING SPATIAL, SPECTRAL, AND DEEP-LEARNING BASED FEATURE FUSION

The invention provides a system (100) for lip-sync authenticity detection using spatial, spectral, and deep-learning based feature fusion. The system acquires an input video, extracts frames at a fixed temporal rate, and constructs paired lip-region frames through face and lip detection, followed by standardized cropping. Spatial forensic features including correlation, pixel-difference, and edge-density values, together with spectral features derived from a two-dimensional Fast Fourier Transform, are computed from the paired frames. A convolutional neural network with five convolution blocks and an attention mechanism processes the same paired frame to generate deep feature scores and classification probabilities. A feature-fusion engine combines handcrafted and deep-learning features to compute a Lip-Audio Synchronization Consistency Index (LASCI) for determining REAL_SYNC or FAKE_DESYNC classifications. An explainability module generates heatmaps, FFT maps, feature plots, and reasoning outputs to support transparent forensic assessment of lip-sync integrity.

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